SYSTEM ARCHITECTURE & TECHNICAL DOCUMENTATION

DRIVEANDGO

Complete UML, ERD & Data Flow Diagram (DFD) Reference for System Documentation & Thesis Defense

Document Type: Technical Architecture Appendix

Scope: Use Case · Class · Sequence · ERD · Data Dictionary · DFD (L0–L2)

Notation: UML 2.5 · Crow's Foot ERD · Gane & Sarson / Yourdon DFD

Revision: v1.0 — Formal Manuscript Edition

23 DATABASE TABLES	4 ACTOR ROLES	3 DFD LEVELS	11 DOMAIN CLASSES

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This appendix accompanies the DriveAndGo capstone manuscript and is formatted for direct print or PDF export via the browser's Print function (Ctrl+P). Page numbers shown above are indicative of manuscript pagination; enable "Headers and footers" in the print dialog for automatic numbering.

SECTION 01

1. Executive System Overview

DriveAndGo is an enterprise-grade car rental and fleet management ecosystem composed of three integrated tiers, engineered to support end-to-end vehicle reservation, chauffeur dispatch, telematics monitoring, and digital settlement.

1.1 Admin Operations Center (Desktop & Web)

A WinForms / WebView2 hybrid React dashboard supporting real-time telematics visualization, fleet assignment, revenue tracking, OCR-based document verification, dynamic pricing, and automated compliance monitoring for the administrative and fleet-management workforce.

1.2 Customer & Driver Mobile Application

A React Native mobile client serving both renters and chauffeurs, providing vehicle discovery, instant reservation, digital key check-in, in-trip GPS routing, driver bidding, split billing among co-travelers, and digital payment receipt issuance.

1.3 Backend Web API Core

A .NET 10 ASP.NET Web API backed by a PostgreSQL relational database, exposing real-time communication through SignalR hubs, securing access via JWT authentication, and persisting documents and media through a multi-tier cloud object storage layer.

1.4 Primary Actors

ACTOR	ROLE IN THE SYSTEM
Admin / Fleet Manager	Oversees fleet configuration, approves bookings and payments, assigns chauffeurs, monitors telematics and compliance, reviews analytics.
Customer / Renter	Browses the fleet, books vehicles with or without a driver, applies promo codes, settles payment, rates completed trips.
Chauffeur / Driver	Submits bids for chauffeured trips, receives dispatch assignments, reports incidents, communicates with customers and admin.
External Systems	Payment gateways (GCash / Maya / Bank) and IoT telematics / weather sensors that exchange data with the core platform.

SECTION 02

2. UML Use Case Diagram

The diagram below details the functional boundaries of DriveAndGo and the interactions of its four primary actors — Admin / Fleet Manager, Customer / Renter, Chauffeur / Driver, and External Systems (Payment Gateways, Weather & GPS Telematics) — against sixteen core use cases (UC-01 through UC-16).

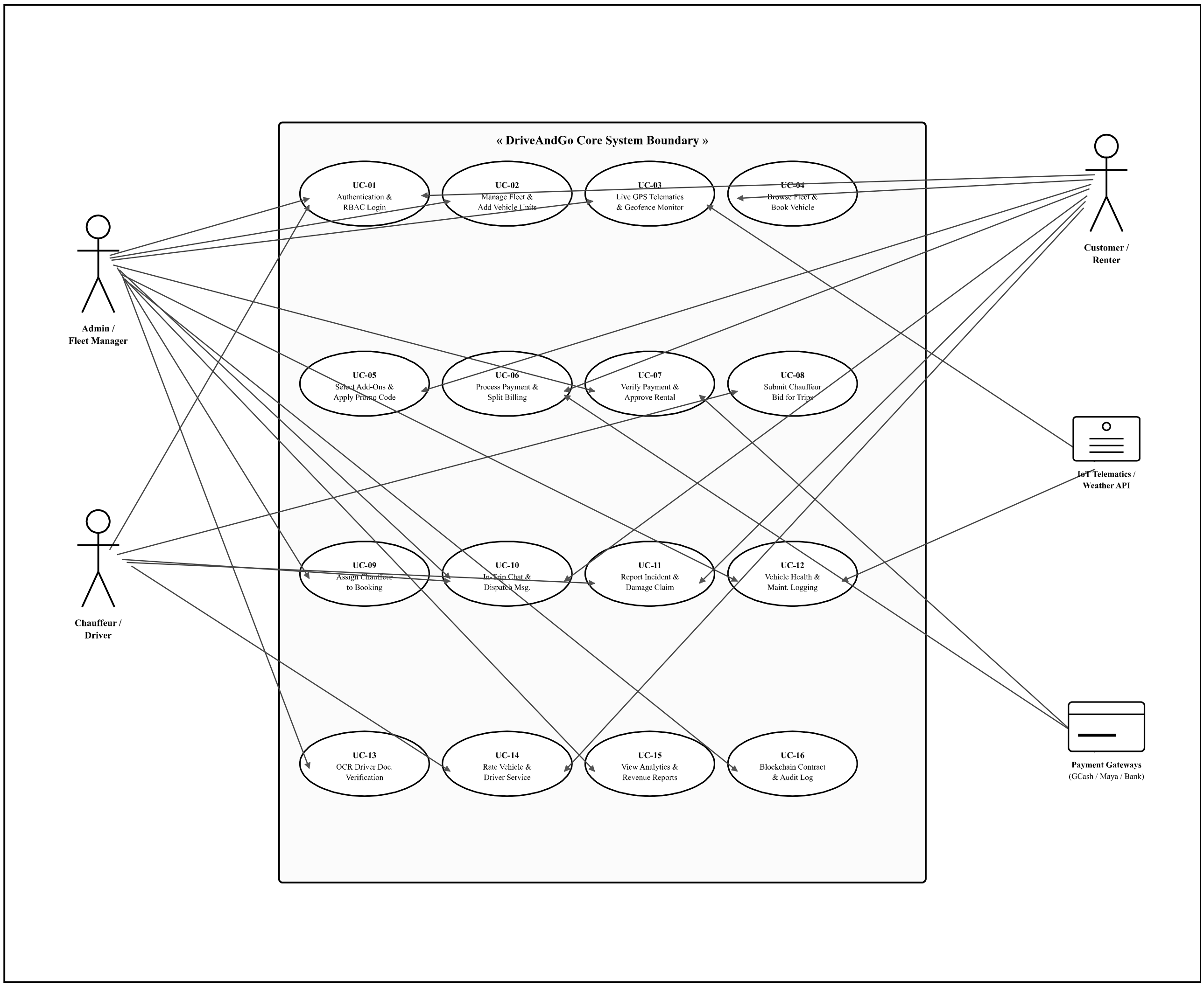

Figure 2.1 — UML Use Case Diagram: Actors, System Boundary, and Sixteen Core Use Cases (UC-01–UC-16)

SECTION 03

3. UML Class Diagram (Domain Model & OOP Architecture)

The domain model below reflects the Object-Oriented entities and their business-logic associations as implemented in the C# .NET 10 API backend, including inheritance (`User → Driver`), one-to-many bookings, and transactional/audit relationships.

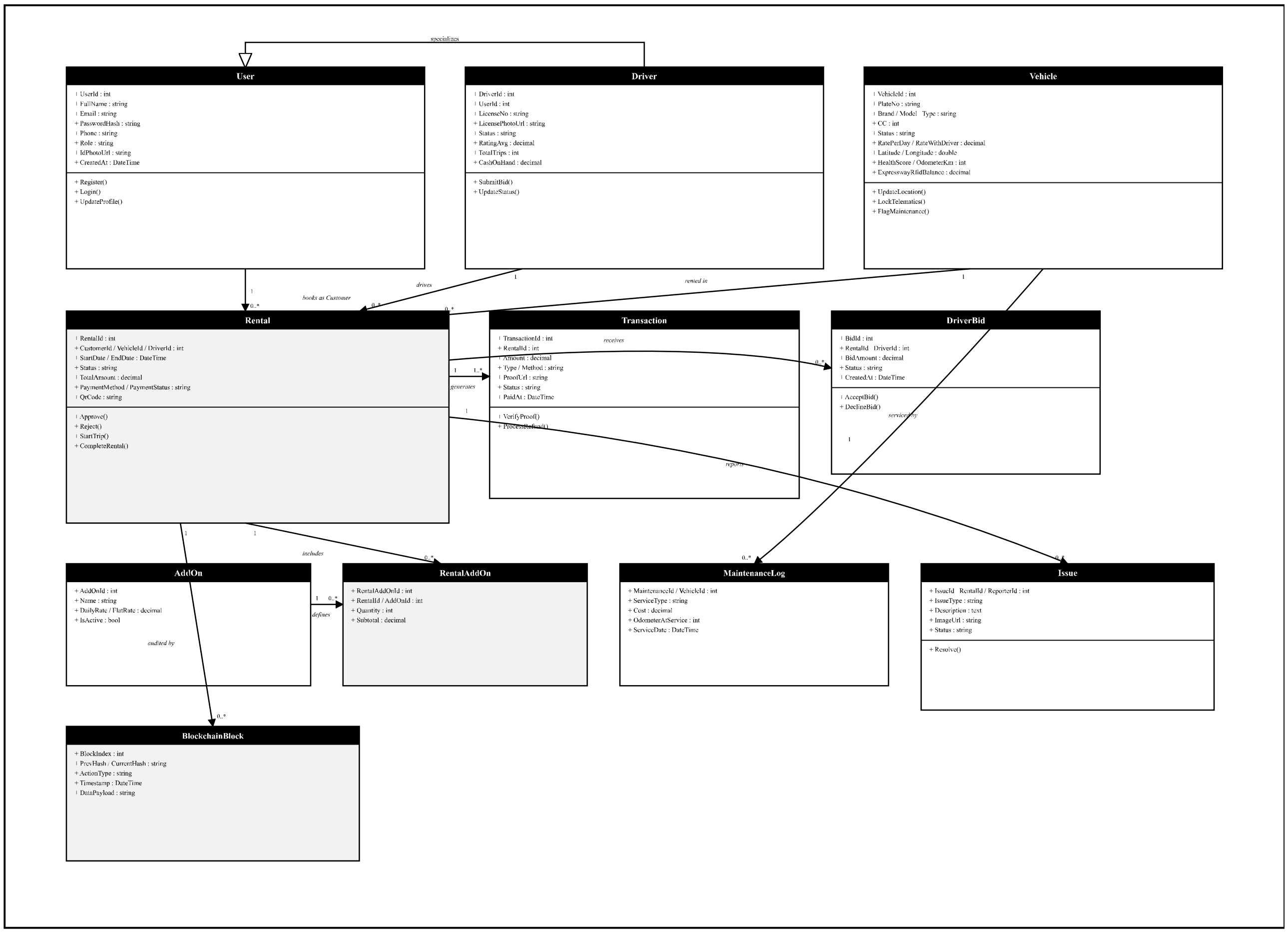

Figure 3.1 — UML Class Diagram: Domain Entities, Attributes, Methods, and Cardinalities

SECTION 04

4. UML Sequence Diagram

End-to-end Booking, Chauffeur Bidding, Telematics, and Settlement flow across seven lifelines.

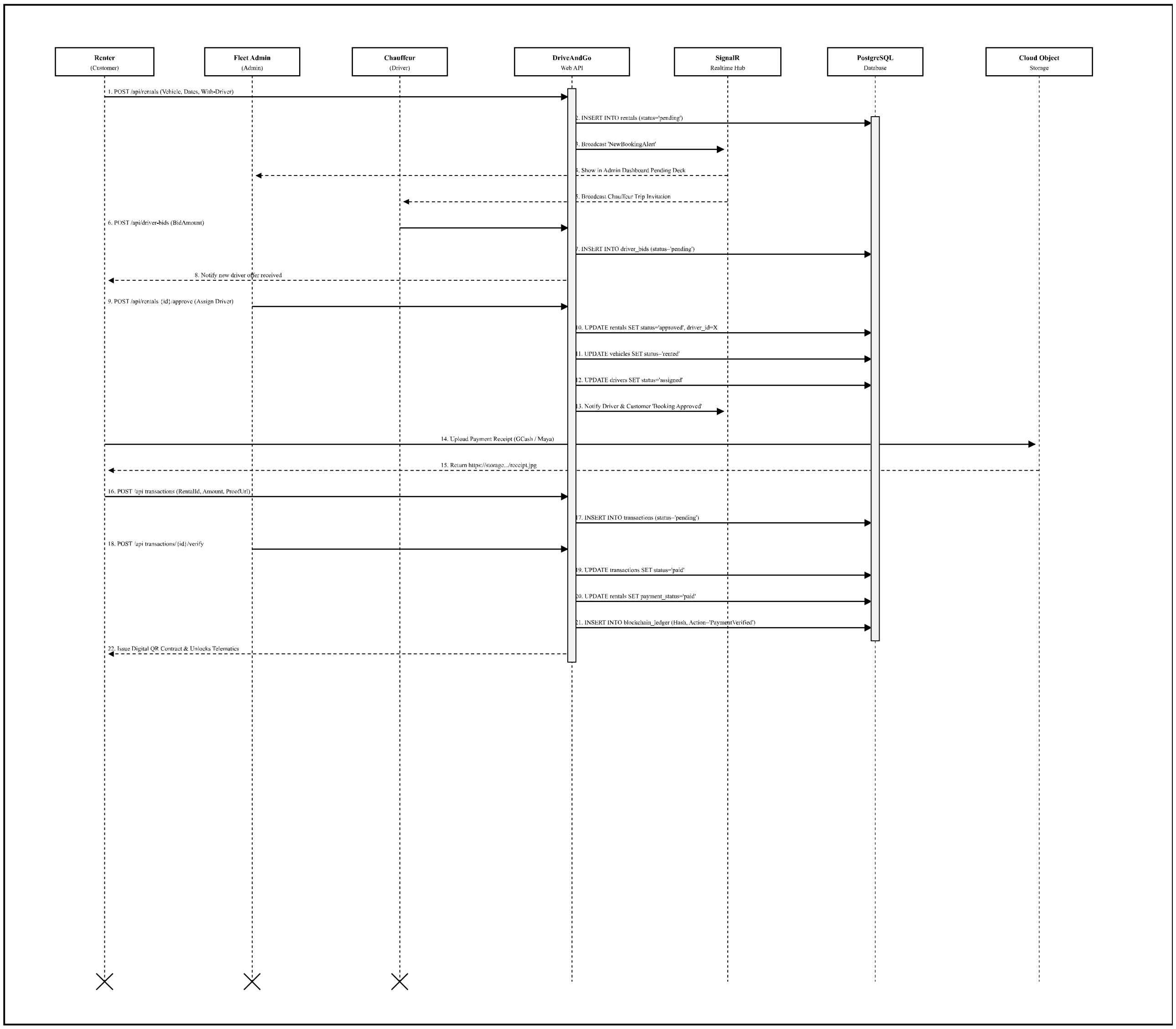

Figure 4.1 — UML Sequence Diagram: Booking, Chauffeur Bidding, Telematics Unlock & Settlement (Messages 1–22)

SECTION 05

5. Complete Entity-Relationship Diagram (ERD)

Crow's Foot notation across all 23 database tables. The fourteen core transactional entities are rendered with full attribute schemas; nine satellite/reference tables are shown in compact form, linked to their primary parent entity.

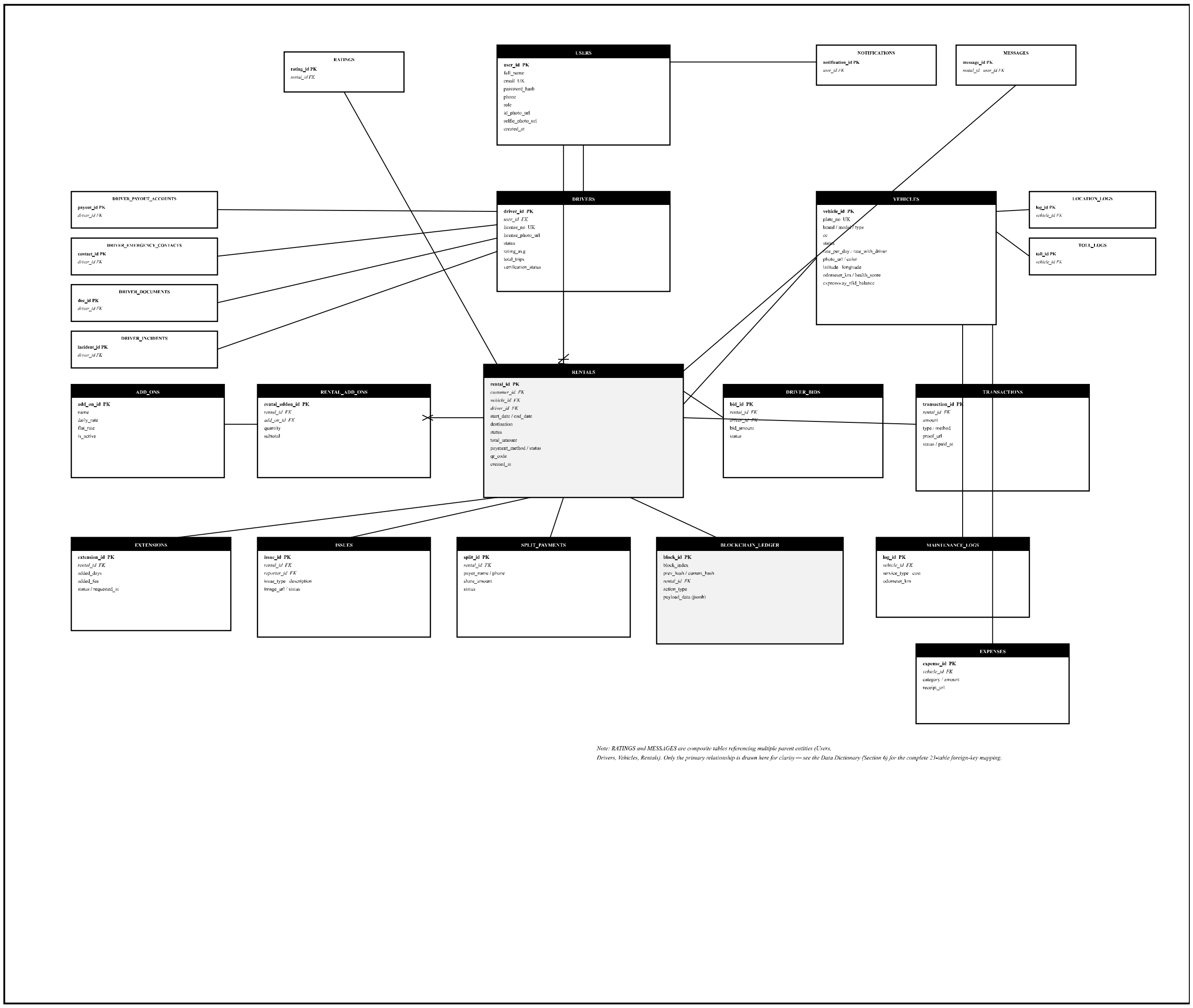

Figure 5.1 — Crow's Foot Entity-Relationship Diagram: 23 Database Tables

SECTION 06

6. Comprehensive Data Dictionary

Table 6.1 — Core table definitions: primary keys, foreign keys, key columns, and descriptions.

TABLE	PRIMARY KEY	FOREIGN KEYS	KEY COLUMNS & TYPES	DESCRIPTION
users	user_id (SERIAL)	None	email (VARCHAR), role (VARCHAR), password_hash (TEXT)	System actors (Customers, Drivers, Admins, SuperAdmins).
drivers	driver_id (SERIAL)	user_id → users	license_no (VARCHAR), status (VARCHAR), rating_avg (NUMERIC)	Professional chauffeur workforce profile & performance metrics.
driver_documents	doc_id (SERIAL)	driver_id → drivers	doc_type (VARCHAR), file_url (TEXT), status (VARCHAR)	LTO license, NBI, police, drug-test compliance vault.
vehicles	vehicle_id (SERIAL)	None	plate_no (VARCHAR), brand (VARCHAR), rate_per_day (NUMERIC), status (VARCHAR)	Fleet repository incl. IoT telematics coordinates, speed, health score.
rentals	rental_id (SERIAL)	customer_id, vehicle_id, driver_id	status (VARCHAR), total_amount (NUMERIC), start_date (TIMESTAMPTZ)	Core business contract binding renter, vehicle, driver, and dates.
transactions	transaction_id (SERIAL)	rental_id → rentals	amount (NUMERIC), method (VARCHAR), proof_url (TEXT), status (VARCHAR)	Financial audit records for payments, refunds, deposits, extensions.
driver_bids	bid_id (SERIAL)	rental_id, driver_id	bid_amount (NUMERIC), status (VARCHAR)	Auction bidding records submitted by drivers for with-driver bookings.
add_ons	add_on_id (SERIAL)	None	name (VARCHAR), daily_rate (NUMERIC), flat_rate (NUMERIC)	Accessories catalog (GPS Navigator, Baby Seat, Luggage Rack).
rental_add_ons	rental_addon_id	rental_id, add_on_id	quantity (INT), subtotal (NUMERIC)	Line items of add-ons selected per rental reservation.
promo_codes	promo_id (SERIAL)	None	code (VARCHAR), discount_type (VARCHAR), discount_value (NUMERIC)	Discount coupons and promotional campaigns.

TABLE	PRIMARY KEY	FOREIGN KEYS	KEY COLUMNS & TYPES	DESCRIPTION
maintenance_logs	log_id (SERIAL)	vehicle_id → vehicles	service_type (VARCHAR), cost (NUMERIC), odometer_km (INT)	Scheduled maintenance, oil changes, brake pads, and repairs.
expenses	expense_id (SERIAL)	vehicle_id → vehicles	category (VARCHAR), amount (NUMERIC), receipt_url (TEXT)	Fleet operating costs (Fuel, Cleaning, Tolls, Emergency Repairs).
split_payments	split_id (SERIAL)	rental_id → rentals	payer_name (VARCHAR), share_amount (NUMERIC), status (VARCHAR)	Peer-to-peer split billing records for group travel expenses.
issues	issue_id (SERIAL)	rental_id, reporter_id	issue_type (VARCHAR), description (TEXT), image_url (TEXT)	Road accidents, vehicle damage, flat tire, or engine issues.
blockchain_ledger	block_id (SERIAL)	rental_id → rentals	prev_hash (TEXT), current_hash (TEXT), action_type (VARCHAR)	Immutable SHA-256 cryptographic chain preventing booking fraud.

The remaining nine satellite tables — notifications, messages, ratings, driver_payout_accounts, driver_emergency_contacts, driver_documents, driver_incidents, location_logs, and toll_logs — complete the full 23-table schema and are represented in compact form in Figure 5.1.

SECTION 07.1

7.1 DFD Level 0: Context Diagram

Establishes the highest-level view of information exchange between DriveAndGo and all external entities.

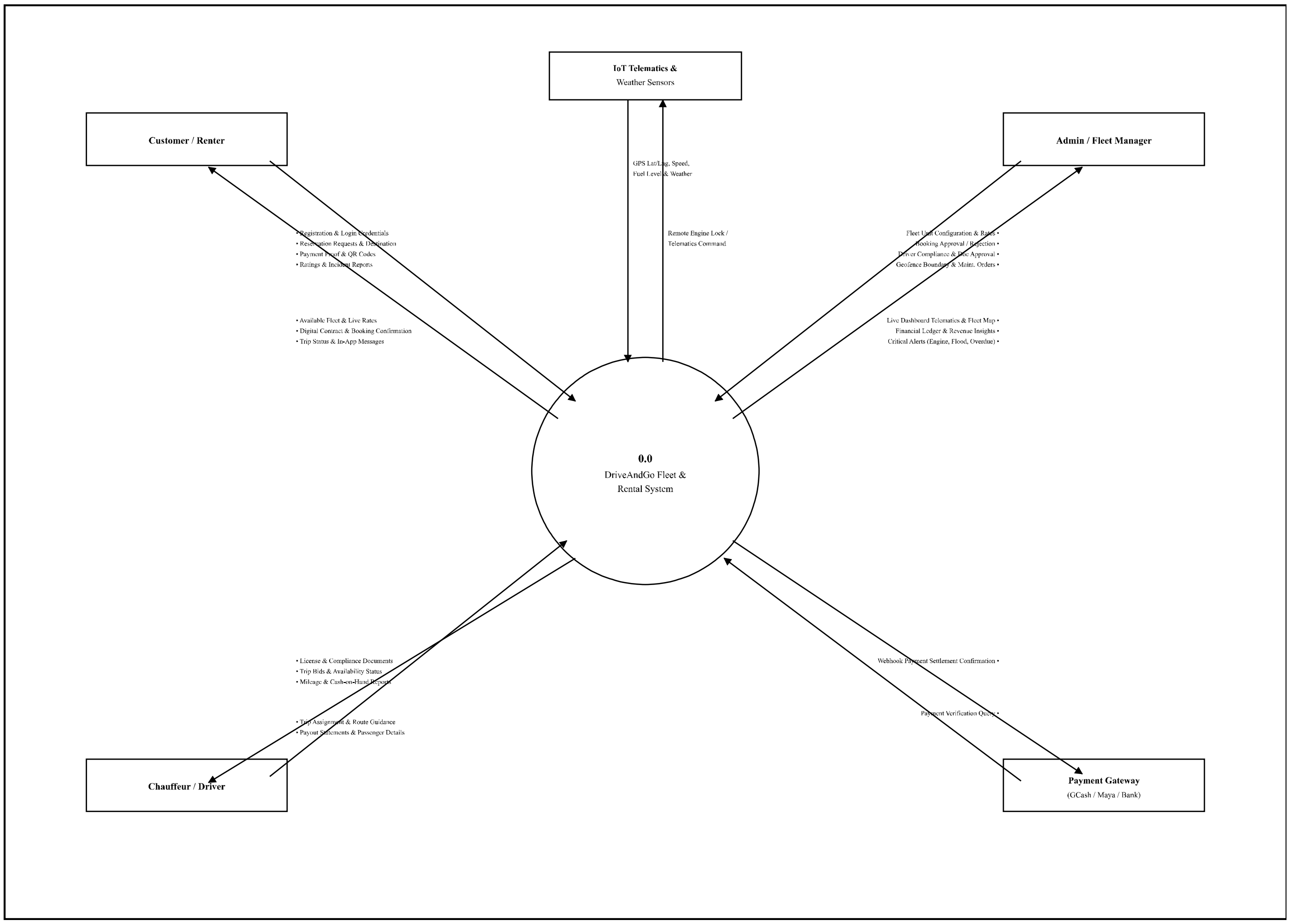

Figure 7.1 — DFD Level 0: Context Diagram

SECTION 07.2

7.2 DFD Level 1: System Decomposition Diagram

Decomposes the core system into seven major functional processes and six data stores (D1–D6).

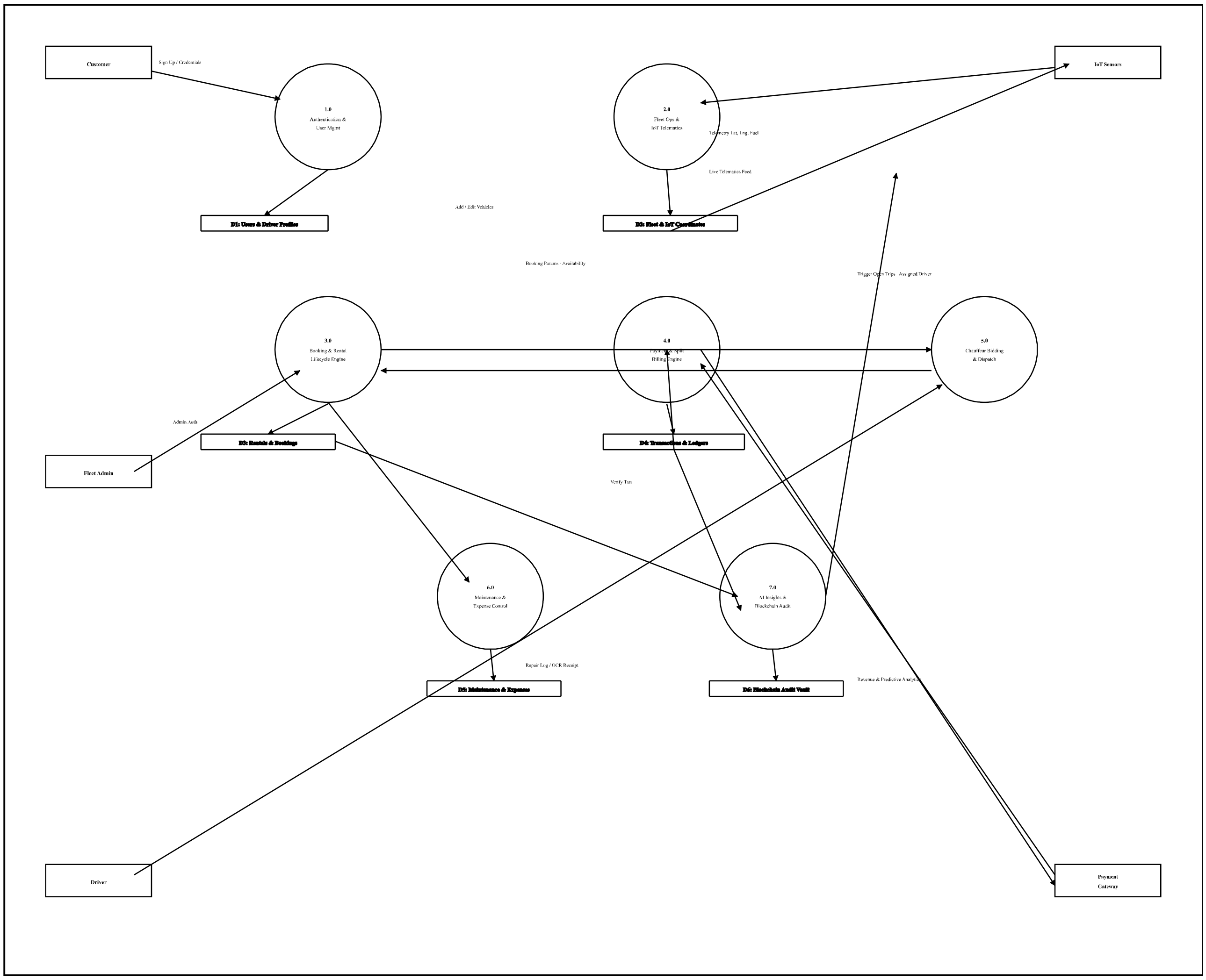

Figure 7.2 — DFD Level 1: System Decomposition (Processes 1.0–7.0, Data Stores D1–D6)

SECTION 07.3

7.3 DFD Level 2: Booking & Payment Subsystem

A microscopic trace of reservation requests flowing through pricing, discount calculation, proof verification, and immutable ledger anchoring.

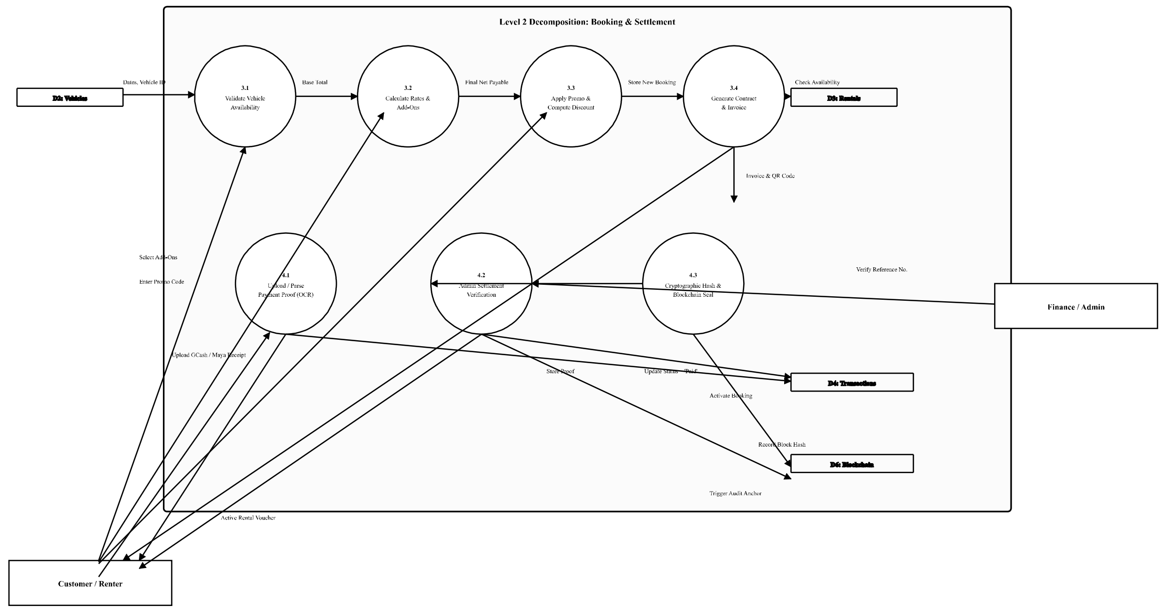

Figure 7.3 — DFD Level 2: Booking & Payment Subsystem Decomposition

SECTION 08

8. Summary & Capstone / Documentation Readiness

ARTIFACT	COVERAGE
UML Use Case	Fully outlines user interactions and roles of all actors, with system boundaries matching academic software-engineering standards.
UML Class Diagram	Maps clean 1-to-Many and Object-Oriented inheritance between Users, Drivers, Fleet Vehicles, Contracts, and Bids.
UML Sequence Diagram	Demonstrates synchronous & asynchronous events across WebSockets (SignalR), Controllers, and Cloud Storage.
ERD & Data Dictionary	Crow's Foot ERD detailing all 23 database tables with data types, constraints, and relationships.
DFD Levels 0, 1, 2	Gane & Sarson / Yourdon notation mapping the inputs, outputs, processes, and data stores of the entire platform.

— End of Technical Architecture Appendix —